Practice Exam Questions

June 14 - July 5, 2019

1. Consider a linear bivariate regression model with an intercept: $y_i = \beta_0 + \beta_1 x_i + u_i$.
   (a) Write the model in vector notation. Be clear about the dimension of the vectors, etc.
   (b) Show that the linear conditional mean model $x'\beta$ minimizes the expected quadratic error loss, and therefore, β can be interpreted as the best linear predictor. Obtain a formula for β based on the moments of x_i and y_i.
   (c) Using the method of moments, show that under some assumptions, we can obtain an unbiased estimator for β. State the assumptions.
   (d) Assume that the variance of the error term is constant, say $\mathrm{V}(u_i) = \sigma^2$. Derive the variance of $\hat{\beta}_0$ and $\hat{\beta}_1$.
2. Consider again the linear bivariate regression model: $y_i = \beta_0 + \beta_1 x_i + u_i$.
   (a) Assuming that x_i is a continuous independent variable, define β_1 as a function of the conditional mean model and provide an interpretation.
   (b) What is the definition of β_0?
   (c) Assuming that $x_i = \{0, 1\}$ is a binary discrete variable, define β_1 as a function of the conditional mean model and provide an interpretation.
3. The demand for Colombia coffee in the US is assumed to be determined by the following function
   $$q = \beta_0 + \beta_1 p_c + \beta_2 p_b + \beta_3 p_d + u$$
   where q denotes imports of Colombian coffee in the US, p_c denotes the price of Colombian coffee, p_b stands for the price of Brazilian coffee, and p_d is the price of Italian coffee. All variables are in natural log.
   (a) Suppose you estimate the equation using OLS considering $n = 200$ observations,
   $$\begin{aligned} \hat{E}(q|p_c, p_b, p_d) &= 2.837 - 1.481 p_c + 1.181 p_b + 0.186 p_d \\ & \quad\ (2.000) \quad (0.987) \quad (0.650) \quad\ (0.134) \end{aligned}$$

with the sum of square residuals $RSS = 0.4277$. Also,

$$\begin{aligned}\hat{E}(q|p_c, p_b, p_d) &= \underset{(0.820)}{-0.738} - \underset{(1.011)}{p_c} + \underset{(0.155)}{0.199 p_d}\end{aligned}$$

with $RSS = 0.4485$. Test the following hypothesis using a 5 % level of significance: (i) $H_0 : \beta_1 = -1$; (ii) $H_0 : \beta_2 = 0$. Then test the null hypothesis that $H_0 : \beta_1 = -1$ and $\beta_2 = 0$. What do you conclude? (Critical values: 1.96 for the t distribution; 2.99 for the F distribution; 5.99 for the χ^2_2).

(b) A referee argues that the error u might be heteroskedastic. Explain how you would answer her/his question. Be explicit about the test, null hypothesis and alternative hypothesis. If the R-square of a regression of $\hat{u}^2$ on covariates is 0.1, would you worry about heteroskedastic errors?

(c) Suppose you reject the null hypothesis in (b) and you note that p_c is 10 times larger than p_b in one month (but not in the other months). Explain how you would estimate the model. Explain also what are the practical caveats associated with the implementation of your method(s).

4. Consider the following linear model:

$$E(y|x_1, x_2) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2, \tag{1}$$

where x_1 and x_2 have zero means.

(a) Write an equation for the dependent variable y and state the assumptions for the error term u as implied by the conditional mean model (1).

(b) Find the partial effect of $E(y|x_1, x_2)$ with respect to x_1. What is the interpretation of β_1?

(c) Construct an elasticity with respect to a change in x_j.

(d) Explain how you will estimate the elasticity you constructed before and state the assumptions needed for obtaining an unbiased and consistent result.

5. Consider a location model,

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} + \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$$

where the *iid* variable $u_1 \sim u_2(0, \sigma^2)$. Show that pooling the data and estimating $\beta = \beta_1 = \beta_2$ reduces the variability of $\hat{\beta}_j$.

6. Show that $M \equiv I - X(X'X)^{-1}X'$ is an orthogonal projection matrix. Show that the residual vector $\hat{u} = Mu$ is orthogonal to the fitted vector $\hat{y}$.

7. Consider,

$$y = X_1\beta_1 + X_2\beta_2 + u$$

where $Euu' = \sigma^2 I$. Suppose you have

$$\tilde{\beta}_1 = (X_1'X_1)^{-1}X_1'y$$

$$\hat{\beta}_1 = (X_1'M_2X_1)^{-1}X_1'M_2y$$

   (a) Explain the difference between the estimators.
   (b) Define M_2 and check if it is idempotent. Are $\tilde{\beta}_1$ and $\hat{\beta}_1$ biased? Explain.
   (c) Find the expression for $\hat{\beta}_1 = \tilde{\beta}_1 - B\hat{\beta}_2$.
   (d) Show that an application of the FWL theorem gives an estimator that is numerically equivalent to $\hat{\beta}_1$ and the standard error of $\hat{\beta}_1$.
   (e) If the error u is not homocedastic, would the covariance between $\tilde{\beta}_1$ and $B\hat{\beta}_2$ be zero? Why is this result important?

8. Let A be a nonsingular matrix of the form,

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \tag{2}$$

   (a) Derive the partitioned inverse formula

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}^{-1} = \begin{bmatrix} W^{-1} & -W_{12}^{-1}A_{12}A_{22}^{-1} \\ -A_{22}^{-1}A_{21}W^{-1} & A_{22}^{-1} + A_{22}^{-1}A_{21}W^{-1}A_{12}A_{22}^{-1} \end{bmatrix}$$

   where $W = A_{11} - A_{12}A_{22}^{-1}A_{21}$ (HINT: solve for the partitioned elements of B from the formula $AB = I$)
   (b) Use the previous formulae to derive the OLS estimator for $\hat{\beta}_1$ constructively from the fitting formula

$$\hat{\beta} = (X'X)^{-1}X'y = \begin{bmatrix} X_1'X_1 & X_1'X_2 \\ X_2'X_1 & X_2'X_2 \end{bmatrix}^{-1} \begin{bmatrix} X_1'y \\ X_2'y \end{bmatrix}$$

   (c) Find the conditional variance of $\hat{\beta}_1$ assuming that the error term $u \equiv y - X\beta$ is normally distributed, say $u \sim \mathcal{N}(0, \sigma^2)$

9. Considering the following regression model,

$$y = X\beta + Z\alpha + u$$

with $X \perp Z$. Suppose you estimate the model with using least squares methods. Show that (i) $\hat{\alpha}$ does not depend on X and (ii) $Cov(\hat{\beta}, \hat{\alpha}) = 0$.

10. Let $\hat{\beta}$ be the OLS estimator and $\tilde{\beta}$ any other linear unbiased estimator. Show that if $E(u|X) = 0$ and $E(uu'|X) = \sigma^2 I$ in a linear regression model, then the OLS estimator is efficient e.g. $V(\tilde{\beta}|X) \geq V(\hat{\beta}|X)$ or

$$V(\tilde{\beta}|X) - V(\hat{\beta}|X) \text{ is positive semi-definite}$$

11. State, demonstrate mathematically, and explain the practical implications of the Frish-Waugh-Lovell Theorem.

12. Consider the following model

$$y = X\beta + u$$

where the mean of the error term is zero and the covariance between x_i and u_i is zero.

(a) Assuming that the error term u_i is correlated with u_j, how you would estimate the parameter β and the standard error of $\hat{\beta}$? Why?

(b) Explain what are the issues of estimating the standard error of $\hat{\beta}$ by OLS.

(c) Suppose now that the $Cov(u_j, u_i) = 0$ for all i and j and the conditional variance is $E(uu'|X) \equiv \Omega$. Obtain the best linear unbiased estimator and its variance. Show that the error term of the transformed model has constant variance.

(d) Explain the steps needed to implement the method suggested in (c).

(e) Show that the difference between the variance of the OLS estimator and the variance of the estimator you proposed in (c) is positive-semi definite.

13. Consider a random sample $Z_1, Z_2, \ldots, Z_m$ distributed independently from a uniform distribution $\mathcal{U}[0, \theta]$.

(a) Obtain a consistent estimator of θ.

(b) Show that the estimator is consistent, explaining the role of the assumptions needed to establish consistency.

14. State the conditions under which the OLS estimator for a linear model $y = X\beta + u$ is consistent and asymptotically normal. Prove the consistency of the estimator and obtain the variance of the asymptotic distribution.

15. Consider the first equation in a linear system of equations,

$$y_1 = \alpha_1 Y_2 + x'\beta + u_1 \tag{3}$$

where Y_2 is the (suspected) endogenous variable. Show that the OLS estimator of the following equation,

$$y_1 = \alpha_1 Y_2 + x'\beta + \gamma\hat{r} + v_1 \tag{4}$$

is identical to the 2SLS estimator. ($\hat{r}$ denotes the residuals from $Y_2 = z_i'\pi + r_i$, where the variable z's are exogenous).

16. Considering the following regression model,

$$y_i = x_i'\beta + \epsilon_i \tag{5}$$

with $\mathbb{E}(\epsilon_i|x_i) = 0$ and known $\mathbb{E}(\epsilon_i^2|X_i) = \sigma^2$.

   (a) Show that the generalized least squares estimator can be written as an instrumental variable estimator.

   (b) Write the expression for the instrumental variable.

17. (a) Show that the following assumption implies orthogonality between the random variable in x_i and the random variable $y_i - x_i'\beta$:

$$\mathbb{E}[x_i(y_i - x_i'\beta_0)] = 0 \tag{6}$$

   Give additional conditions on the joint distribution of x_i and y_i that make β_0 the unique solution to

$$\mathbb{E}[x_i(y_i - x_i'\beta)] = 0 \tag{7}$$

   (b) Show that the OLS estimator $\hat{\beta}$ is analogous to β_0 in the sense that it is the unique solution to

$$\frac{1}{n}\sum x_i(y_i - x_i'\beta) = 0 \tag{8}$$

   where sample moments have replaced population moments. This is an example of a method-of-moments estimator, which constructs parameter estimators that equate empirical moments with population counterparts.

18. Consider a regression model

$$y_i = x_i'\beta + \epsilon_i \qquad i = 1, \ldots, n$$

in which ϵ_i is a latent random variable that is correlated with the explanatory variable in x_i. Let the elements of x_i include the constant 1. Assume *iid* sampling and that

$$\begin{aligned} \mathbb{E}[\epsilon_i] &= 0 \\ Var[\epsilon_i] &= \sigma_0^2 \end{aligned}$$

and

$$\mathbb{E}[x_i x_i'] = D_{xx}$$

where D_{xx} is a positive-definite matrix. In addition, let $Cov[x_i, \epsilon_i] = \rho_0$.

(a) Consider the sample moments of (x_i, y_i): $\bar{x} \equiv (1/n)\sum_{i=1}^{n} x_i$, $\bar{y} \equiv (1/n)\sum_{i=1}^{n} y_i$,$(1/n)X'X$, $(1/n)y'y$, and $(1/n)X'y$. Find their expected values in terms of the unknown parameters

(b) Suppose that there are K instrumental variables, $z_{ik}(k = 1, \ldots, K)$ that are uncorrelated with ϵ_i,

$$Cov[z_i, \epsilon_i] = 0$$

yet correlated with x_i,

$$\mathbb{E}[z_i x_i'] = D_{zx}$$

where D_{zx} is also nonsingular. Find the expected values of the additional sample moments $(1/n)\sum_{n=1}^{n} z_i$, $(1/n)Z'Z$, and $(1/n)Z'X$ in terms of the unknown parameters.

(c) By equating all the first and second sample moments above to their population values, find a method-of-moments estimator for all of the unknown paramters. Show in particular that the estimator of β_0 is the IV estimator $(Z'X)^{-1}Z'y$

19. Find the maximum likelihood estimator for the unknown parameter θ considering the following distributions:

(a) $f(x, \theta) = \theta \exp\{-\theta x\}$, with $x \geq 0$ and $\theta > 0$

(b) $f(x, \theta) = \theta c^{\theta} x^{-(\theta+1)}$, with $x \geq c$, $\theta > 0$

(c) $f(x, \theta) = x\theta^{-2} \exp\{-x^2/2\sigma^2\}$, $x > 0, \theta > 0$

(d) $f(x, \theta) = \sigma^{-1} \exp\{-(x - \mu)/\sigma\}$, $x \geq \mu$ where $\theta = (\mu, \sigma)$

(e) $x_1, \ldots, x_n$ from a distribution $u[\theta - 1/2, \theta + 1/2]$.

(f) $x_1, \ldots, x_n$ from $\mathcal{N}(\mu_1, \sigma^2)$ and $Y_1, \ldots, Y_n$ *iid* from $\mathcal{N}(\mu_2, \sigma^2)$. Note that $\theta = (\mu_1, \mu_2, \sigma^2)$

20. Suppose X is a continuous random variable with probability density function $f(x, \theta_0)$. Assume continuous second order derivatives. Show that $\mathbb{E}(\partial l(\theta_0|x)/\partial\theta) = 0$. Assuming that $Var(\partial l(\theta_0|x)/\partial\theta)$ $\exists$, show that:

$$\mathbb{E}\left(\frac{\partial^2 l(\theta_0|x)}{\partial\theta\partial\theta'}\right) = -Var\left(\frac{\partial l(\theta_0|x)}{\partial\theta}\right)$$